

Helix Thready — Brand Assets (Launcher Icon, Logo, Slogan)

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Rev	Date	Author	Change
1	2026-07-21	swarm (design)	Initial complete draft: icon concept, construction, detail tiers, light/dark, OS export matrix, org logo usage, footer slogan + heart
2	2026-07-22	swarm (design · Pass 3)	Depth pass: numeric icon geometry spec (grid/safe-zones/coil growth, §3.1); concrete per-platform manifests — <code>manifest.webmanifest</code> , favicon <code><head></code> , Android <code>ic_launcher.xml</code> + adaptive/ monochrome, iOS <code>Contents.json</code> , HarmonyOS layered JSON (§5.1); per-surface slogan-placement matrix (§8.1); corrected heart-color provenance to the verified <code>--accent-ink</code> token + exact i18n keys (<code>footer.made</code> / <code>footer.by</code> / <code>a11y.love</code>)

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1. Requirements (verbatim intent)

From the original request (§ “Launcher icon / main logo design”) `[OPERATOR]`:

- Create the launcher icon with **OpenDesign** `[CONSTITUTION §11.4.162]`.
- **No letters** — the OS prints the app title, so the mark must not contain any text.
- Reflect the project’s **central theme, purpose, and a creative idea**.
- The **Helix element** (the defining element of the Helix Development logo) **must be part of** the product icon.
- Vector master; exported in **all major sizes/formats for all supported OSes**.
- Legible at **very large and very small** sizes.
- Provide **multiple versions** (more / less detail) used per context.
- Provide **light and dark** versions with **transparent background**.
- The **Helix Development org logo** is still used in footers/branding so it is clear the software is part of Helix Development.

2. The mark: reading Logo.png

`assets/Logo.png` `[VERIFIED — inspected]` is a **spiral** — an unrolling nautilus/thread coil — that reads simultaneously as:

- a **helix / spiral** (the Helix Development defining element, satisfying “helix must be part of the icon”), and
- a **thread** unspooling into a coil (the literal meaning of *Thready* — reading threads of posts), and
- a **chameleon-tail / shell** silhouette (adaptive, organic — matching the product that adapts to many content types).

Color reading (informs the theme in `design-system.md`):

Region	Color (approx.)	Role
Outer sweep, left	Chartreuse / lime green	Primary brand — <code>--brand helix-green #B6E376 [VERIFIED base]</code> (Logo.png green-region eyedrop <code>#BAE448</code> , $n \approx 1.06M$, corroborates)
Flowing to right	Mint / soft teal	Secondary — <code>--brand-2 teal #ABDDC9 [VERIFIED -- eyedrop mean of Logo.png mint region, n=618,886; median #B7EBD6]</code>
Spiral negative space	White	The coil / thread path

The mark is **already letter-free** and **already carries the Helix spiral** — so the product icon is a refinement of this master, not a new invention.

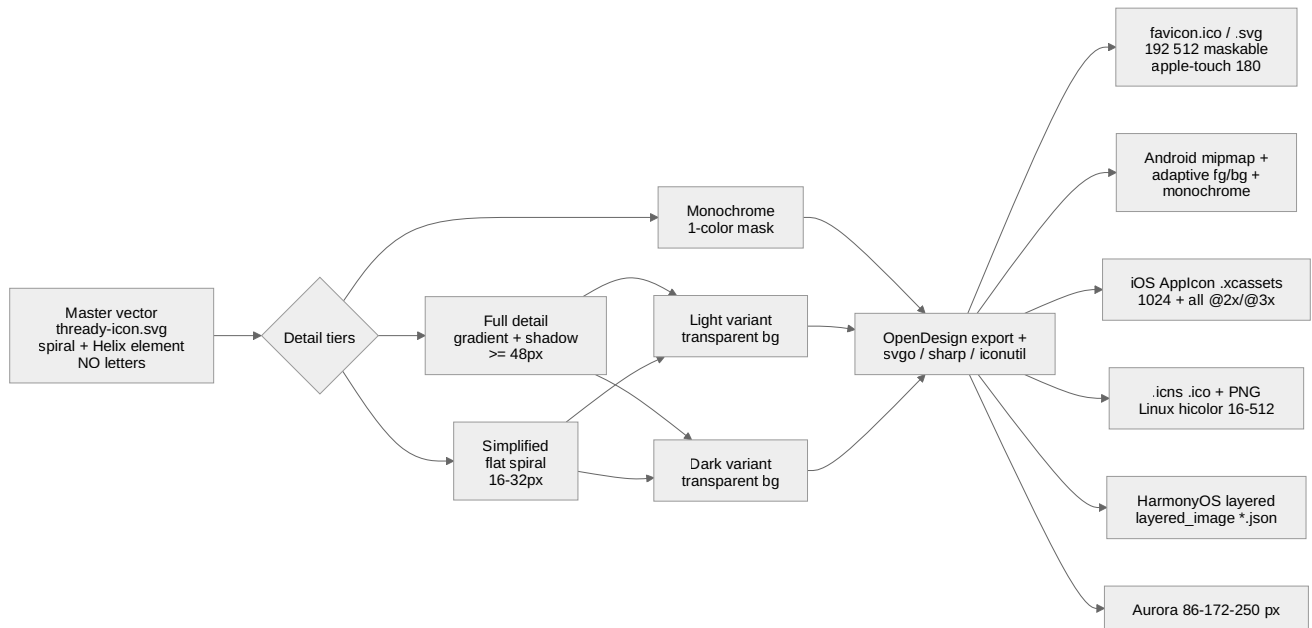
3. Launcher/Icon concept & construction

Concept: “*The thread becomes a helix.*” A single ribbon starts as a straight thread at the outer edge and coils inward into a tight helix/spiral eye — the reading loop of a thread rendered as the Helix element. Letter-free by construction.

Construction rules (authored in OpenDesign, master as SVG):

- **Grid:** designed on a 1024×1024 master with a safe area. Two safe zones are cut:
 - a **circular** safe zone (Android adaptive, iOS squircle, macOS/Windows rounded) — the spiral eye is centered so no coil is clipped by platform masking;
 - a **full-bleed** zone for maskable/adaptive backgrounds.
- **Geometry:** the ribbon width, coil count and eye position are proportion-locked to the master; the spiral uses a logarithmic (golden-ratio-ish) growth so it stays balanced when simplified.
- **Fill:** a chartreuse→teal gradient (`--brand #B6E376` → `--brand-2 #ABDDC9`), both eyedrop-captured from (Logo.png) on the ribbon; white/transparent negative space for the coil path. A **flat** (non-gradient) fill is the small-size fallback.

- **No letters, no wordmark inside the icon.** The wordmark “Thready” is a *separate* lockup used only in headers/marketing, never inside the launcher icon.



Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `diagrams/icon-export-pipeline.mmd`.

Explanation (for readers/models that cannot see the diagram). A single master vector (`thready-icon.svg`) — the spiral/Helix element, with no letters — is the only hand-authored source. From it three **detail tiers** are derived: a full-detail version (gradient + subtle elevation, used at $\geq 48\text{px}$), a simplified flat spiral (16–32px, where gradients and shadows muddy), and a monochrome one-color mask (for tinted/adaptive contexts). Each tier is produced in a **light** and a **dark** variant, both on transparent backgrounds so they compose over any UI. All variants then pass through the **export pipeline** (OpenDesign export plus `svgo` for SVG optimization, `sharp` for raster sizes, `iconutil` / `png2icns` for macOS, and platform packagers). The pipeline emits every platform bundle: web favicons + PWA maskable icons + apple-touch, Android mipmaps with an adaptive foreground/background pair and a monochrome layer, the iOS `AppIcon` asset catalog, the desktop `.icns` / `.ico` / PNG set (Linux hicolor 16–512), the HarmonyOS layered image, and the Aurora sizes. Because the master is a single vector, every downstream size stays perfectly consistent.

3.1 Numeric geometry spec

So the master is reproducible (not “draw a nice spiral”), the construction is pinned to numbers on the **1024×1024** master `[DEFAULT — adjustable, pending the OpenDesign authoring pass]`:

Property	Value	Rationale
Master canvas	1024×1024 px	OS-canonical largest source
Keyline safe circle (Android adaptive / iOS squircle)	Ø 768 px, centered (75%)	Android adaptive masks to ~66–75%; the spiral eye sits at center so no coil clips
Full-bleed zone (maskable / adaptive bg)	full 1024, art within Ø 916	maskable icons need art inside the 80% inner circle
Spiral eye center	canvas center (512, 512)	mask-safe under circle and squircle
Ribbon stroke width	96 px at outer turn → 40 px at the eye	tapered ribbon reads as a thread narrowing into the coil
Coil count	2.75 turns (full tier) → 2.0 (simplified) → 1.75 (mono)	fewer turns as size shrinks so it never muddies
Growth	logarithmic, ratio ≈ 1.618 per quarter-turn (golden)	stays balanced when simplified
Clear-space	≥ 128 px (⅛ box ≈ one outer ribbon width)	matches \$10 clear-space rule
Corner radius (adaptive fg)	n/a — art only; the OS applies the mask	never bake platform corners into the art

Fills. Full tier = a linear gradient `--brand #B6E376 → --brand-2 #ABDDC9` along the ribbon's sweep, white/transparent negative space for the coil path. Simplified/mono tiers = a single flat tone (`--accent #446E12` light, `--brand #B6E376` dark) so the mark survives at 16–32 px and in tinted Android-monochrome / notification slots. **No letters** are ever inside any tier (the OS prints the title). The `--brand-2` endpoint is the eyedrop-captured `#ABDDC9` `[VERIFIED — closes THREADY-DES-01]`.

4. Detail tiers & light/dark variants

Tier	Used at	Contents	Fallback of
Full	≥ 48px (launcher, splash, store)	Gradient ribbon, soft inner shadow, full coil count	—
Simplified	16–32px (favicon, tab, list rows)	Flat single-tone ribbon, reduced coil count, thicker stroke	Full
Monochrome	Adaptive/tinted (Android monochrome, notification, watch)	1-color silhouette of the spiral	Simplified

Light vs. dark (transparent bg, per requirement):

- **Light variant:** ribbon in `--brand` → `--brand-2` gradient (or `--accent #446E12` for the flat/mono tone), tuned for light UIs and light home screens.
- **Dark variant:** ribbon brightened to the logo lime `#B6E376` (the verified dark accent) so it holds on dark surfaces; coil negative space stays transparent (reads as the dark surface).
- Selection at runtime follows the same three mechanisms as the theme (`( prefers-color-scheme / [data-theme] / .dark )`) for in-app marks; OS launcher icons use the platform's own light/dark icon slots where available (iOS 18 dark/tinted, Android monochrome).

5. OS / platform export matrix

All exported from the single master `[DEFAULT — adjustable]`. Sizes are the platform-canonical set.

Web / PWA

Asset	Sizes
<code>favicon.svg</code> (scalable) + <code>favicon.ico</code>	16, 32, 48
PWA <code>icon</code> (manifest)	192, 512
PWA <code>maskable</code>	192, 512 (full-bleed safe zone)
<code>apple-touch-icon.png</code>	180

Android (`res/mipmap-*` + adaptive)

Asset	Densities
Legacy <code>ic_launcher.png</code>	mdpi 48, hdpi 72, xhdpi 96, xxhdpi 144, xxxhdpi 192
Adaptive <code>ic_launcher_foreground</code> + <code>_background</code>	108dp @ all densities (72px safe circle)
<code>ic_launcher_monochrome</code> (Android 13+ themed)	108dp
Play Store	512×512

iOS (`Assets.xcassets/AppIcon.appiconset`)

Asset	Sizes
App icon (single-size, Xcode 14+)	1024 (system downscales)
Legacy full set (if targeting older)	20/29/40/58/60/76/80/87/120/152/167/180
Dark + tinted (iOS 18)	1024 each

Desktop

OS	Format	Sizes
macOS (Tauri)	<code>.icns</code>	16, 32, 64, 128, 256, 512, 1024 (@1x/@2x)
Windows (Tauri)	<code>.ico</code>	16, 24, 32, 48, 64, 128, 256
Linux (Tauri)	PNG hicolor	16, 22, 24, 32, 48, 64, 128, 256, 512 + scalable SVG

HarmonyOS / Aurora `[GAP: 8.5 – native clients scaffold]`

OS	Format	Notes
HarmonyOS	layered image (<code>foreground</code> + <code>background</code> PNG + <code>layered_image</code> JSON)	via ArkTS client (scaffold)
Aurora (auroraos.ru)	PNG	86, 108, 128, 172, 250 (per Sailfish/Aurora density buckets) <code>[RESEARCH – verify at integration]</code>

5.1 Concrete platform manifests

The export pipeline emits these files verbatim (not just sizes) so each client wires the icon with no guesswork `[DEFAULT – adjustable]`:

Web — `manifest.webmanifest` + `<head>`:

```
{
  "name": "Thready", "short_name": "Thready",
  "theme_color": "#020817", "background_color": "#ffffff", "display": "standalone",
  "icons": [
    { "src": "/icons/icon-192.png", "sizes": "192x192", "type": "image/png" },
    { "src": "/icons/icon-512.png", "sizes": "512x512", "type": "image/png" },
    { "src": "/icons/maskable-192.png", "sizes": "192x192", "type": "image/png", "purpose": "maskable" },
    { "src": "/icons/maskable-512.png", "sizes": "512x512", "type": "image/png", "purpose": "maskable" }
  ]
}
```

```
<!-- self-hosted, no external CDN (CSP hygiene) -->
<link rel="icon" href="/icons/favicon.svg" type="image/svg+xml">
<link rel="icon" href="/icons/favicon.ico" sizes="16x16 32x32 48x48">
<link rel="apple-touch-icon" href="/icons/apple-touch-icon.png" sizes="180x180">
<link rel="manifest" href="/manifest.webmanifest">
<meta name="theme-color" content="#ffffff" media="(prefers-color-scheme: light)">
<meta name="theme-color" content="#020817" media="(prefers-color-scheme: dark)">
```

Android — `res/mipmap-anydpi-v26/ic_launcher.xml` (adaptive + Android-13 monochrome):

```
<adaptive-icon xmlns:android="http://schemas.android.com/apk/res/android">
  <background android:drawable="@color/ic_launcher_background"/> <!-- #FFFFFF light /
    handled by themed -->
  <foreground android:drawable="@drawable/ic_launcher_foreground"/> <!-- spiral, art within
    66dp safe -->
  <monochrome android:drawable="@drawable/ic_launcher_monochrome"/> <!-- 1-color spiral
    (themed icons) -->
</adaptive-icon>
```

iOS — `Assets.xcassets/AppIcon.appiconset/Contents.json` (single-size + dark + tinted, iOS 18):

```
{ "images": [
  { "idiom": "universal", "platform": "ios", "size": "1024x1024", "filename": "icon-1024.png" },
  { "idiom": "universal", "platform": "ios", "size": "1024x1024", "filename": "icon-1024-
    dark.png",
    "appearances": [ { "appearance": "luminosity", "value": "dark" } ] },
  { "idiom": "universal", "platform": "ios", "size": "1024x1024", "filename": "icon-1024-
    tinted.png",
    "appearances": [ { "appearance": "luminosity", "value": "tinted" } ] }
], "info": { "author": "opendesign-export", "version": 1 } }
```

HarmonyOS — `resources/base/media/layered_image.json` (ArkTS client, `[GAP: 8.5 scaffold]`):

```
{ "layered-image": { "background": "$media:ic_launcher_background",
  "foreground": "$media:ic_launcher_foreground" } }
```

Aurora consumes plain PNGs at the density buckets in §5 `[OPEN: THREADY-DES-05 — verify at integration]`. Because all of these reference art derived from the **one** master, a master change re-emits every file through Docs Chain (§6).

6. Export pipeline

```
# thready-icons.export.yaml [DEFAULT – adjustable]
master: assets/brand/thready-icon.svg           # single hand-authored vector, no letters
tiers: { full: ">=48", simplified: "16-32", monochrome: "adaptive" }
variants: [light, dark]                         # transparent background
targets:
  web: { favicon: [16,32,48], svg: true, pwa: [192,512], maskable: [192,512], appleTouch:
        180 }
  android: { mipmap: [48,72,96,144,192], adaptive: true, monochrome: true, playstore: 512 }
  ios: { appicon: 1024, dark: true, tinted: true }
  macos: { icns: [16,32,64,128,256,512,1024] }
  windows: { ico: [16,24,32,48,64,128,256] }
  linux: { hicolor: [16,22,24,32,48,64,128,256,512], svg: true }
  harmonyos: { layered: true }                  # via ArkTS client [GAP: 8.5]
  aurora: { png: [86,108,128,172,250] }         # [RESEARCH]
tools: [opendesign-export, svgo, sharp, png2icns, icotool]
verify:
  - render each target size and assert legibility (no clipped coil, >=3:1 mark/bg where placed)
  - snapshot into the visual-regression bank (ScreenDiff) # [GAP: 9.3]
```

The pipeline is wired into **Docs Chain** so a change to the master re-exports every size `[CONSTITUTION §11.4.65]` and the design↔project hooks required by the request (“all design materials ... completely connected ... through Docs Chain”).

7. Helix Development org logo (attribution)

The Helix Development org logo (green helix, `assets/logos/helix-development-logo.png` in `design_system [VERIFIED]`) remains the **attribution** mark. It appears — never removed by white-labeling — in:

- app **footers** (alongside the slogan, §8),
- generated-document footers (§9),
- the “About”/legal surface.

Per §8.3 white-labeling (see `theming.md`): an Account may override the *product* logo/colors/slogan, but the **Helix Development attribution persists** in footers.

8. The footer slogan & heart glyph

Required string `[OPERATOR]`: “Made with love by Helix Development”, where “love” is replaced by a heart glyph in a proper color.

The in-house `design_system` **already ships this exact pattern** in `reference.footer.component.ts` [VERIFIED — inspected source], and Thready reuses it:

- The heart is the canonical **lucide Heart SVG** (`fill=currentColor`), tinted by a brand token, with `[attr.aria-label]="i18n.t('ally.love')"` so the **accessible name reads “Made with love by Helix Development”**.
- The text around it is localized via the exact shipped i18n keys `footer.made` ... heart ... `footer.by` (`[VERIFIED — {{ i18n.t('footer.made') }} / {{ i18n.t('footer.by') }} in the component]`), so the slogan translates (en/ru/sr-Cyrl) without hard-coding.
- **Ordered visual fallbacks** (the head script toggles `html.no-svg`, applied via `:host-context(html.no-svg)` in the shipped component): lucide SVG → `U+2665` (♥) glyph → literal word “love”. This guarantees the slogan renders in any environment (including the TUI and plain-text/email exports).

```
<!-- Thready footer slogan (adapted from design_system reference.footer) [VERIFIED pattern] -->
<p class="made-with">
  Made with
  <span class="heart" role="img" aria-label="love" title="love">
    <svg class="heart-svg" viewBox="0 0 24 24" width="18" height="18"
      fill="currentColor" stroke="none" aria-hidden="true" focusable="false">
      <path d="M19 14c1.49-1.46 3-3.21 3-5.5A5.5 5.5 0 0 0 16.5 3c-1.76 0-3 .5-4.5
        2-1.5-1.5-2.74-2-4.5-2A5.5 5.5 0 0 0 2 8.5c0 2.3 1.5 4.05 3 5.5l7 7Z"/>
    </svg>
    <span class="heart-glyph" aria-hidden="true">&#9829;</span> <!-- ♥ fallback -->
    <span class="heart-word" aria-hidden="true">love</span> <!-- text fallback -->
  </span>
  by Helix Development
</p>
```

```
.heart { color: var(--ds-heart); display:inline-flex; align-items:center; line-height:0; }
.heart-glyph, .heart-word { display:none; }
html.no-svg .heart-svg { display:none; } html.no-svg .heart-glyph { display:inline; }
```

Heart “proper color”. The shipped precedent tints the heart with `var(--accent-ink)` — the accessible accent-ink token — so it is AA-legible in light and dark [VERIFIED — `.heart { color: var(--accent-ink); }` in `reference.footer.component.ts`]. Thready exposes this as a `--ds-heart` token that defaults to that accent-ink (in the Thready theme, `--accent` already resolves to the AA-pinned `#446E12` light / `#B6E376` dark). Whether the operator prefers a **classic love-red** instead is [OPEN: THREADY-DES-03]; if red is chosen, keep it distinct from `--danger` (error) and ensure the mark is ≥ 3:1 as a meaningful non-text graphic. The heart is decorative (its meaning is carried by the accessible name “love”), so a red does not need text-contrast — but a legible mark is still required.

Surface variants of the slogan:

- **Web/Desktop/Mobile:** the SVG heart (above).

- **TUI/CLI:** the `U+2665` (♥) glyph, tinted with the Lipgloss `Accent` (or `Danger` -red) style — e.g. `Made with ♥ by Helix Development`.
- **Generated Markdown/PDF/HTML:** ♥ glyph (matches this doc's own footer).

8.1 Per/surface slogan placement

The slogan and attribution appear on **every** surface; placement and glyph differ by medium but the accessible string is always “Made with love by Helix Development”. This is the locked contract the white-label cannot remove (§7, [theming §3](#)):

Surface	Location	Heart rendering	Attribution logo
Web / Desktop portal	<code>.ds-footer</code> app-shell footer (§ wireframes §3.1)	lucide SVG, <code>--ds-heart</code>	Helix Development mark beside slogan
Mobile (More tab / About)	Settings › About + app footer	SVG (Compose/SwiftUI vector)	mark in About
TUI	bottom rail under the live pane (§ wireframes §5)	♥ glyph, Lipgloss <code>Accent</code>	text “Helix Development”
CLI	<code>thready version</code> / <code>--help</code> epilogue	♥ glyph (or <code>love</code> if <code>--no-color</code> /no-UTF8)	text
Generated docs (PDF/HTML/MD)	document footer , every page	♥ glyph	Helix Development attribution logo (§9)
Login / splash	under the wordmark (“read your threads, smarter”)	SVG	product logo (Account-branded) above
Launcher icon	never — the icon is letter-free by construction	—	—

The one place the slogan/heart must **not** appear is inside the launcher icon (§1: “no letters”).

9. Brand on generated documents

Per §8.3 and the request (“Adding logo into header and footers of all documents + slogan”): every generated artifact (research docs, books, reports) carries, styled by the design system via **Docs Chain**:

- **Header:** the effective **product** logo (Account-branded if white-labeled, else Thready) + document title.

- **Footer:** the **Helix Development** attribution logo + the “Made with ♥ by Helix Development” slogan + document metadata (revision, date).
- Typography, code/syntax-highlight, diagram and callout styles are the design-system tokens (\$ design-system.md), so generated documents read as one system with the apps [CONSTITUTION \$11.4.65].

10. ClearSpace, misuse, and file inventory

Clear-space: minimum padding around the icon = the ribbon width ($\approx 1/8$ of the icon box). Do not place other marks inside the clear-space.

Misuse (do not): add letters inside the icon; recolor outside `--brand / --brand-2 /mono`; skew or rotate the spiral; drop the transparent background; use the full-detail tier below 48px; use the product icon where the Helix Development attribution logo is required.

File inventory (proposed, under `assets/brand/` [DEFAULT – adjustable]):

```
assets/brand/
  thready-icon.svg          # master (no letters)
  thready-icon.mono.svg
  thready-wordmark.svg      # "Thready" lockup (headers/marketing only)
  light/ dark/              # per-variant exports (transparent bg)
  export/                   # generated OS bundles (web, android, ios, macos, windows,
linux, harmonyos, aurora)
  helix-development-logo.svg # attribution (from design_system)
```

11. Gaps & open items

- [GAP: 8.1 design_system] — the Thready brand assets live beside the design-system brand assets; add `thready` theme + Thready icon set to the published package (workable item THREADY-DES-DS-01).
- [GAP: 8.5 helix_shims / HarmonyOS+Aurora] — HarmonyOS layered icon + Aurora sizes are produced now but consumed by native clients that are still scaffolds; do not claim those launchers ship.
- [CLOSED: THREADY-DES-01] — `--brand-2` teal captured from a formal `assets/Logo.png` eyedrop (mint region mean `#ABDDC9`, $n = 618,886$; dark median `#B7EBD6`), replacing the `#7AA590` stand-in; method in design-system §3.2. Re-confirm with the design-system’s own eyedrop tool at integration.
- [OPEN: THREADY-DES-03] — heart color: brand accent (default) vs. love-red.
- [OPEN: THREADY-DES-05] — Aurora density buckets [RESEARCH] must be verified against current Aurora OS packaging docs at integration.